

Tuned NRTE full-pipeline failure categories by dataset size

Patent-only

	Material only	Geometry only	Material + geometry	CodeV error	Total failure
SMOTE	4.82%	0.00%	0.86%	12.92%	18.60%
STaSy	27.87%	0.01%	58.06%	12.36%	98.30%
TabDDPM	35.03%	0.06%	64.17%	0.74%	100.00%
TabSyn	35.00%	0.01%	59.72%	4.70%	99.43%

250k

	Material only	Geometry only	Material + geometry	CodeV error	Total failure
	0.24%	0.00%	0.00%	1.33%	1.57%
	18.65%	0.02%	7.72%	15.08%	41.47%
	25.16%	0.08%	73.11%	1.65%	100.00%
	11.75%	0.00%	2.48%	8.29%	22.52%

500k

	Material only	Geometry only	Material + geometry	CodeV error	Total failure
SMOTE	0.23%	0.00%	0.00%	1.41%	1.64%
STaSy	18.18%	0.00%	6.22%	12.24%	36.64%
TabDDPM	24.10%	0.05%	74.44%	1.41%	100.00%
TabSyn	11.22%	0.00%	2.76%	10.05%	24.03%

Full

	Material only	Geometry only	Material + geometry	CodeV error	Total failure
	0.17%	0.00%	0.00%	1.35%	1.52%
	16.76%	0.00%	7.99%	10.17%	34.92%
	n/a	n/a	n/a	n/a	n/a
	15.75%	0.00%	4.59%	12.51%	32.85%

Fraction of 10k generated samples

1.0

0.8

0.6

0.4

0.2

0.0